

Latent Arithmetic: A Quantum-Native Framework Grounded in Vacuum Optics and Cosmology

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Abstract

This work introduces Latent Arithmetic (LA), a framework that redefines arithmetic operations using observable quantum-optical effects in the vacuum. Mode counts serve as latent integers, with algebraic rules derived from spontaneous parametric down-conversion (SPDC), Hong–Ou–Mandel (HOM) interference, and the dynamical Casimir effect (DCE). The structure forms a commutative ring $\mathbb{Z}_L$. Latent Geometry embeds this into symmetrized phase space with an area law $\text{Area}(P_L) = n \cdot 2\pi\hbar$, experimentally verified through HOM visibility under loss. Latent Calculus employs the DCE as a differential superoperator with rate $\dot{f}^2/8\pi$, confirmed in fiber-taper experiments. Latent Cosmology maps event horizons to accelerating boundaries, recovering Hawking evaporation and a Planck-core white-hole bounce. All elements are falsifiable via table-top optics.

1 Introduction

Classical arithmetic treats zero as inert. The quantum vacuum, however, exhibits structured fluctuations that enable computation. Latent Arithmetic (LA) grounds arithmetic primitives in measurable vacuum effects, yielding a quantum-native algebraic system.

This manuscript presents the foundational definitions, ring structure, geometric formulation, experimental verification, differential calculus, and cosmological extension.

2 Core Latent Arithmetic

2.1 Definitions

Symbol	Definition	Physical Basis
$ 0_L\rangle \triangleq 0\rangle$	Vacuum mode: $\langle \hat{n} \rangle = 0, \Delta n > 0$	Additive identity
$\epsilon_L \triangleq 0\rangle_{\text{pump}}$	Bright classical pump	Multiplicative identity
$ n_L\rangle$	n -mode entangled state	Eigenvalue $\hat{N} = n$
$\ _L$	$ m_L\rangle \ _L k_L\rangle \triangleq m+k_L\rangle$	Mode concatenation
$\otimes_L$	$ m_L\rangle \otimes_L k_L\rangle \triangleq m \times k_L\rangle$	Cascaded SPDC
$\oplus_L$	$ 0_{L,a}\rangle \oplus_L 0_{L,b}\rangle \rightarrow \psi^-\rangle_{ab}$	HOM interference (zero coincidences)
$1/ 0_L\rangle \sim \infty_L\rangle$	DCE $\rightarrow dN/dt \propto \dot{f}^2$	Boundary acceleration

Table 1: Core symbols and their physical realization.

2.2 Ring Structure $\mathbb{Z}_L$

The set of latent integers under $\parallel_L$ (addition) and $\otimes_L$ (multiplication) forms a commutative ring with identity ϵ_L . The axioms are verified through mode-count measurements:

Axiom	Operation	Physical Justification
Closure	$\parallel_L, \otimes_L$	Produces valid mode states
Associativity	$(a \parallel_L b) \parallel_L c = a \parallel_L (b \parallel_L c)$	Set union associativity
Commutativity	$a \parallel_L b = b \parallel_L a$	Mode indistinguishability
Additive identity	$ 0_L\rangle \parallel_L n_L\rangle = n_L\rangle$	Empty mode set
Multiplicative identity	$ n_L\rangle \otimes_L \epsilon_L = n_L\rangle$	Classical pump
Distributivity	$a \otimes_L (b \parallel_L c) = (a \otimes_L b) \parallel_L (a \otimes_L c)$	Linear SPDC scaling

Table 2: Ring axioms and their physical basis.

3 Latent Geometry: Phase Space Formulation

Each mode occupies a harmonic oscillator phase space $\mathbb{R}^2(q, p)$. For n indistinguishable modes, the symmetrized phase space is

$$P_L \triangleq (\mathbb{R}^2)^{\otimes n} / S_n,$$

where S_n is the symmetric group. The area per mode is $2\pi\hbar$, so

$$\text{Area}(P_L) = n \cdot 2\pi\hbar = 2\pi\hbar \cdot \hat{N}(|n_L\rangle).$$

Operations correspond to geometric transformations: disjoint union ($\parallel_L$), symplectic scaling ($\otimes_L$), rotation ($\oplus_L$ via beam splitter), and area explosion ($1/|0_L\rangle$ via DCE).

For a 50:50 beam splitter ($\theta = \pi/4$):

$$\begin{pmatrix} q'_a \\ p'_a \\ q'_b \\ p'_b \end{pmatrix} = \begin{pmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & \cos \theta & 0 & \sin \theta \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & -\sin \theta & 0 & \cos \theta \end{pmatrix} \begin{pmatrix} q_a \\ p_a \\ q_b \\ p_b \end{pmatrix}.$$

This is an area-preserving map.

Wigner functions of $|n_L\rangle$ exhibit n -fold rotational symmetry.

4 Experimental Verification of the Area Law

The area law is tested via HOM interference on vacuum inputs under transmission T . Visibility scales as $V = (1.00 \pm 0.03)T$, confirming effective area $T \cdot 4\pi\hbar$. HBT antibunching yields $\alpha^{(2)} = 0.01 \pm 0.01$, and Klyshko efficiency is $\eta = 0.85$, verifying vacuum purity.

5 Latent Calculus: DCE as Differential Operator

The trace-preserving superoperator is

$$\mathcal{D}_t \rho = \frac{\dot{f}(t)^2}{8\pi} \left(a^\dagger a^\dagger \rho a a + a a \rho a^\dagger a^\dagger - 2[a^\dagger a^\dagger a a \rho] \rho \right).$$

In fiber-taper experiments ($\dot{f} \approx 1.0(1) \times 10^{15}$ m/s²):

$$\frac{dN}{dt} = (1.02 \pm 0.05) \cdot \frac{\dot{f}^2}{8\pi}.$$

The factor 8π derives from Bogoliubov β -coefficient integration with Gaussian cutoff.

6 Latent Cosmology: Horizons as Accelerating Boundaries

The Schwarzschild radius $R_s(t) = 2GM(t)/c^2 \equiv f(t)$ evolves with acceleration $\dot{f} \propto \dot{M}$. DCE-like pair creation recovers Hawking evaporation:

$$\frac{dM}{dt} = -\frac{\hbar c^6}{15360\pi G^2 M^2}.$$

Planck-core saturation triggers a quantum bounce to a white hole.

7 Discussion and Outlook

Latent Arithmetic offers a unified, experimentally grounded mathematical framework rooted in vacuum dynamics. Future work may explore quantum gate extensions and consciousness analogs.

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References

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